

专题：旋转变换在多元 函数积分中的应用

空间中的单位正交变换，旋转变换

设空间线性变换 $T : (x, y, z) \rightarrow (u, v, w)$ 可表为

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

记 $\vec{e}_1 = (a_{11}, a_{12}, a_{13})$, $\vec{e}_2 = (a_{21}, a_{22}, a_{23})$, $\vec{e}_3 = (a_{31}, a_{32}, a_{33})$,

若(1) $|\vec{e}_1|=1, |\vec{e}_2|=1, |\vec{e}_3|=1$; (单位)

(2) $\vec{e}_1, \vec{e}_2, \vec{e}_3$ 互相垂直, 即 $\vec{e}_1 \cdot \vec{e}_2 = \vec{e}_2 \cdot \vec{e}_3 = \vec{e}_1 \cdot \vec{e}_3 = 0$, (正交)

则称变换 T 为**单位正交变换**. 特别, 若 $\vec{e}_1 \times \vec{e}_2 = \vec{e}_3$, 则称 T 为**旋转变换**.

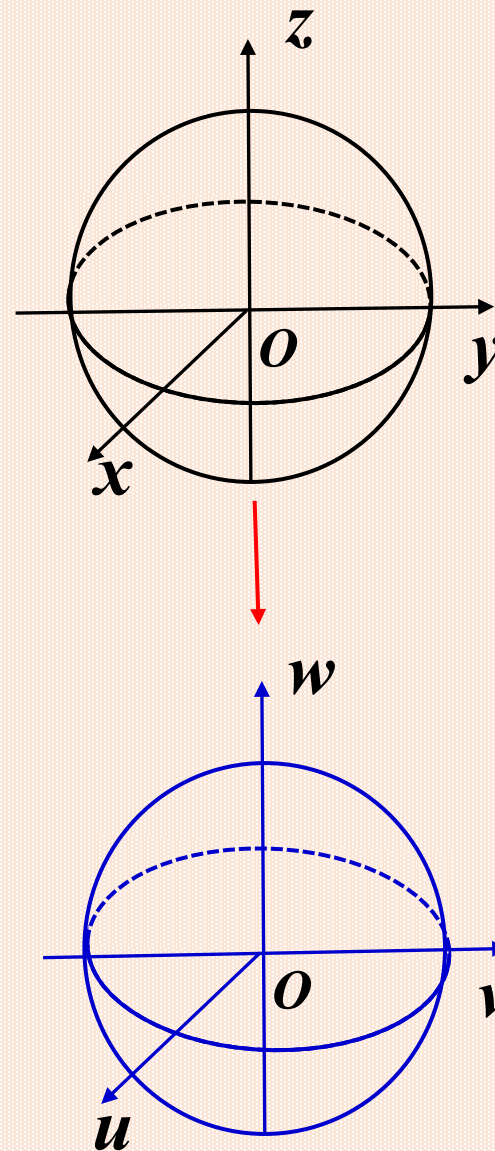
旋转变换的性质

设 $T:(x, y, z) \rightarrow (u, v, w)$ 是旋转变换, 则

(0) $|J| = 1$, 做了变换后区域、曲面、曲线的形状、大小不变.

$$(1) T: \Omega = \{(x, y, z) | x^2 + y^2 + z^2 \leq R^2\} \\ \rightarrow \Omega' = \{(u, v, w) | u^2 + v^2 + w^2 \leq R^2\}$$

$$(2) T: \Sigma = \{(x, y, z) | x^2 + y^2 + z^2 = R^2\} \\ \rightarrow \Sigma' = \{(u, v, w) | u^2 + v^2 + w^2 = R^2\}$$

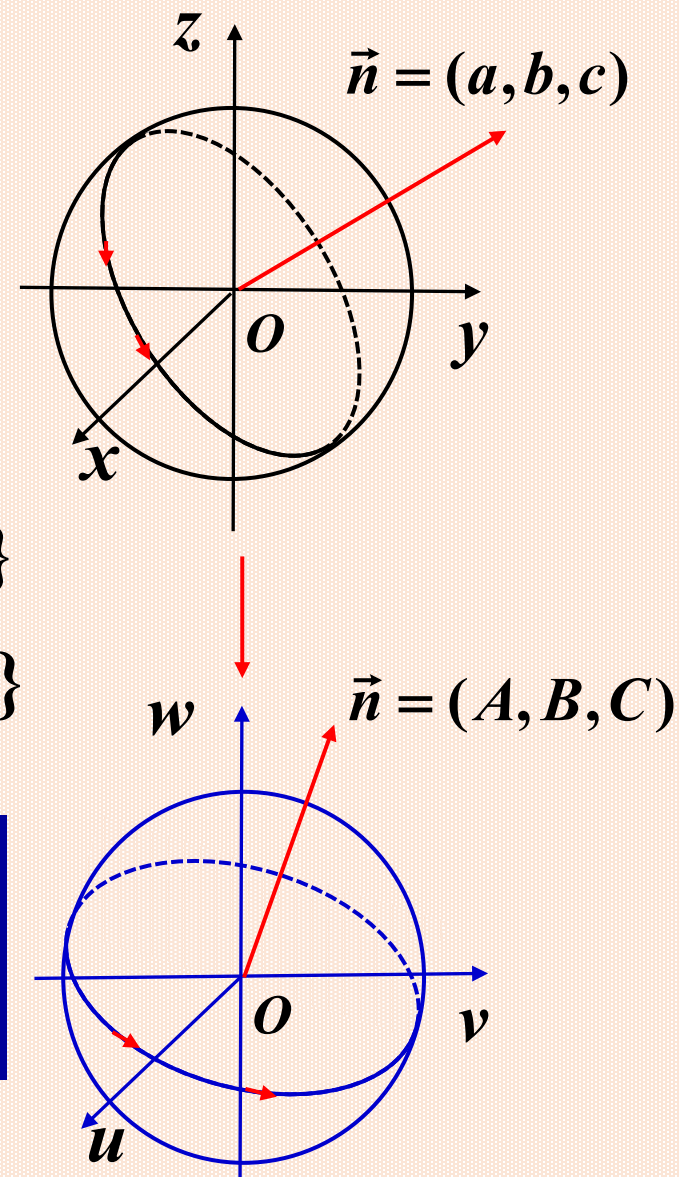


旋转变换的性质

$$(3) \quad T : \Pi = \{(x, y, z) | ax + by + cz = 0\} \\ \rightarrow \Pi' = \{(u, v, w) | Au + Bv + Cw = 0\}$$

$$(4) \quad T : \Gamma = \{(x, y, z) | x^2 + y^2 + z^2 = R^2, ax + by + cz = 0\} \\ \rightarrow \Gamma' = \{(u, v, w) | u^2 + v^2 + w^2 = R^2, Ax + By + Cz = 0\}$$

(5) 若闭曲线 Γ 与 z 轴正向构成右(左)手系,
且 $\vec{e}_3$ 与 z 轴正方向成锐角, 即 $\vec{e}_3 \cdot \vec{k} > 0$,
则闭曲线 Γ' 与 w 轴正向构成右(左)手系.



例1 计算曲线积分 $I = \int_L (y-z)dx + (z-x)dy + (x-y)dz$,

其中 L 是球面 $x^2 + y^2 + z^2 = a^2$ ($a > 0$)与平面 $x + y + z = 0$ 的交线,

从 z 轴正向看去 L 取逆时针方向.

解法一 记 $\vec{e}_3 = \left\{ \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\}$, $\vec{e}_1 = \left\{ \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right\}$,

取 $\vec{e}_2 = \vec{e}_3 \times \vec{e}_1 = \left\{ \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\} \times \left\{ \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right\} = \left\{ \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right\}$,

令 $u = \frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}}$, $v = \frac{x}{\sqrt{6}} + \frac{y}{\sqrt{6}} - \frac{2z}{\sqrt{6}}$, $w = \frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}}$,

则变换 $T : (x, y, z) \rightarrow (u, v, w)$ 为一旋转变换.

变换 T 将 $x^2 + y^2 + z^2 = a^2$ 变为 $u^2 + v^2 + w^2 = a^2$,

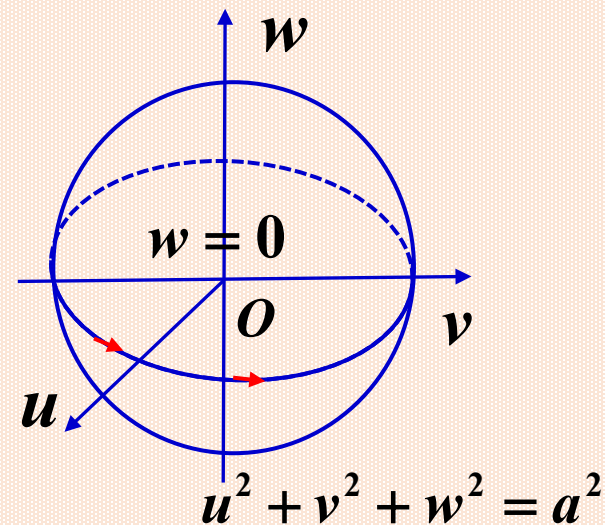
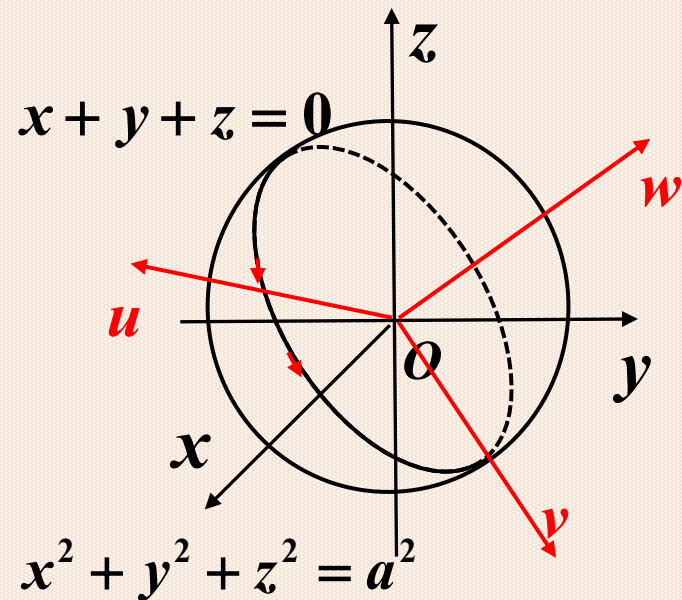
将 $x + y + z = 0$ 变为 $w = 0$. 于是在新坐标系下

L 对应 uov 平面上的圆 $u^2 + v^2 = a^2$. 令

$$u = a \cos t, \quad v = a \sin t, \quad (0 \leq t \leq 2\pi),$$

由 T 的表达式解得

$$x = \frac{u}{\sqrt{2}} + \frac{v}{\sqrt{6}} + \frac{w}{\sqrt{3}}, \quad y = -\frac{u}{\sqrt{2}} + \frac{v}{\sqrt{6}} + \frac{w}{\sqrt{3}}, \quad z = -\frac{2v}{\sqrt{6}} + \frac{w}{\sqrt{3}},$$



则 L 的参数方程为

$$\begin{cases} x = \frac{a}{\sqrt{2}} \cos t + \frac{a}{\sqrt{6}} \sin t, \\ y = -\frac{a}{\sqrt{2}} \cos t + \frac{a}{\sqrt{6}} \sin t, \\ z = -\frac{2a}{\sqrt{6}} \sin t, \end{cases} \quad (t : 0 \rightarrow 2\pi)$$

从而,

$$\begin{aligned} I &= \int_0^{2\pi} [(y(t) - z(t))x'(t) + (z(t) - x(t))y'(t) + (x(t) - y(t))z'(t)] dt \\ &= -2\sqrt{3}a^2\pi \end{aligned}$$

解法二 平面的法向量为 $\vec{n}_1 = \{1, 1, 1\}$, 球面的法向量为 $\vec{n}_2 = \{x, y, z\}$,

于是 L 在点 $(x, y, z) \in L$ 的切向量可取为

$$\vec{n} = \vec{n}_1 \times \vec{n}_2 = \{z - y, x - z, y - x\},$$

从而,

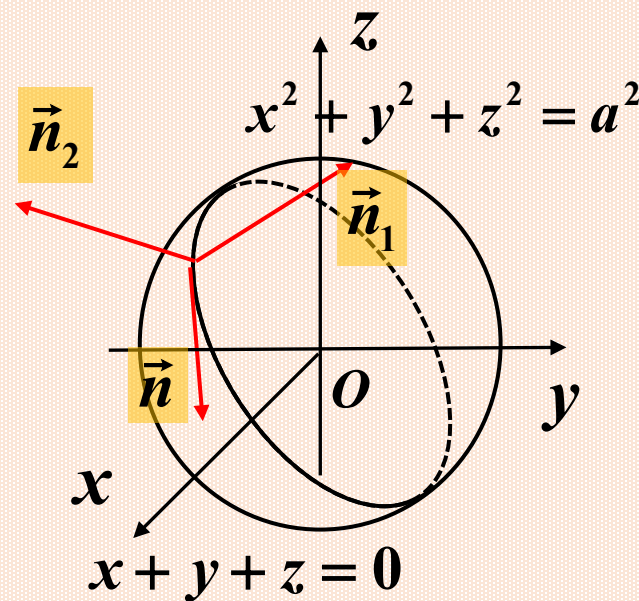
$$(\cos \alpha, \cos \beta, \cos \gamma) = \frac{\{z - y, x - z, y - x\}}{\sqrt{(z - y)^2 + (x - z)^2 + (y - x)^2}},$$

于是,

$$I = \int_L [(y - z) \cos \alpha + (z - x) \cos \beta + (x - y) \cos \gamma] ds$$

$$= - \int_L \sqrt{(y - z)^2 + (z - x)^2 + (x - y)^2} ds$$

$$= - \int_L \sqrt{3(x^2 + y^2 + z^2) - (x + y + z)^2} ds = - \int_L \sqrt{3} a ds = - 2\sqrt{3} a^2 \pi.$$



例2 计算 $I = \iiint_{\Omega} \cos(ax + by + cz) dx dy dz$, 其中 a, b, c 不全为零,

$$\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}.$$

解 作旋转变换 $T : (x, y, z) \rightarrow (u, v, w)$, 使得在新坐标系 (u, v, w) 下

$ax + by + cz = 0$ 对应 $w = 0$. 则 T 形如

$$u = a_1 x + b_1 y + c_1 z, \quad v = a_2 x + b_2 y + c_2 z, \quad w = \frac{ax + by + cz}{\sqrt{a^2 + b^2 + c^2}}.$$

由几何知识知 $|J| = 1$, T 将 Ω 变为 $\Omega' = \{(u, v, w) \mid u^2 + v^2 + w^2 \leq 1\}$.

记 $\mu = \sqrt{a^2 + b^2 + c^2}$, 则 $I = \iiint_{\Omega'} \cos(\mu w) du dv dw.$

设 $u = r \cos \theta$, $v = r \sin \theta$, $w = w$, 则 Ω' 变为

$$\Omega'' = \{(r, \theta, w) \mid -1 \leq w \leq 1, 0 \leq \theta \leq 2\pi, 0 \leq r \leq \sqrt{1-w^2}\},$$

于是,

$$\begin{aligned} I &= \int_{-1}^1 \cos(\mu w) dw \int_0^{2\pi} d\theta \int_0^{\sqrt{1-w^2}} r dr \\ &= \pi \int_{-1}^1 (1-w^2) \cos(\mu w) dw \\ &= \frac{4\pi}{\mu^2} \left(\frac{\sin \mu}{\mu} - \cos \mu \right). \end{aligned}$$

例3 设 $\Omega: x^2 + y^2 + z^2 \leq 1$, 证明 $\frac{4\sqrt[3]{2}}{3}\pi \leq \iiint_{\Omega} \sqrt[3]{x+2y-2z+5} dx dy dz \leq \frac{8}{3}\pi$.

解法一 作旋转变换 $T: (x, y, z) \rightarrow (u, v, w)$, 使得在新坐标系 (u, v, w) 下

$x + 2y - 2z = 0$ 对应 $w = 0$. 则 T 形如

$$u = a_1 x + b_1 y + c_1 z,$$

$$v = a_2 x + b_2 y + c_2 z,$$

$$w = \frac{x + 2y - 2z}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{x}{3} + \frac{2y}{3} - \frac{2z}{3}$$

由几何知识知 $|J| = 1$, T 将 Ω 变为 $\Omega' = \{(u, v, w) | u^2 + v^2 + w^2 \leq 1\}$.

所以,

$$\iiint_{\Omega} \sqrt[3]{x+2y-2z+5} dx dy dz = \iiint_{\Omega'} \sqrt[3]{3w+5} du dv dw.$$

注意到 $-1 \leq w \leq 1$, 所以,

$$\sqrt[3]{3} \leq \sqrt[3]{3w+5} \leq 2,$$

故,

$$\frac{4\sqrt[3]{2}}{3}\pi = \sqrt[3]{2} \iiint_{\Omega'} du dv dw \leq \iiint_{\Omega'} \sqrt[3]{3w+5} du dv dw \leq 2 \iiint_{\Omega'} du dv dw = \frac{8}{3}\pi.$$

解法二先求目标函数 $f(x, y, z) = x + 2y - 2z$ 在 Ω 上的最大最小值.

容易看出 f 在 Ω 内部无驻点, 故最大最小值在边界 $x^2 + y^2 + z^2 = 1$

上取到. 设 $L(x, y, z) = f(x, y, z) + \lambda(x^2 + y^2 + z^2 - 1)$, 解方程组

$$\begin{cases} L'_x = 1 + 2\lambda x = 0 \\ L'_y = 2 + 2\lambda y = 0 \\ L'_z = -2 + 2\lambda z = 0 \\ x^2 + y^2 + z^2 = 1 \end{cases} \quad \text{解得,} \quad (x, y, z) = \left(\pm \frac{1}{3}, \pm \frac{2}{3}, \mp \frac{2}{3} \right)$$

$$f\left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right) = 3, \quad f\left(-\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right) = -3,$$

由于连续函数 $f(x, y, z)$ 在 $\partial\Omega$ 上必有最大值, 最小值.

所以, $\min_{\partial\Omega} f = -3, \max_{\partial\Omega} f = 3$, 这样, $\min_{\Omega} f = -3, \max_{\Omega} f = 3$.

从而,

$$\sqrt[3]{2} = \sqrt[3]{5-3} \leq \sqrt[3]{x+2y-2y+5} \leq \sqrt[3]{5+3} = 2,$$

故,

$$\frac{4\sqrt[3]{2}}{3}\pi = \sqrt[3]{2} \iiint_{\Omega} dx dy dz \leq \iiint_{\Omega} \sqrt[3]{x+2y-2z+5} dx dy dz \leq 2 \iiint_{\Omega} dx dy dz = \frac{8}{3}\pi.$$

练习 设 $f(t)$ 为连续函数, $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

证明
$$\iiint_{\Omega} f(ax + by + cz) dx dy dz = \pi \int_{-1}^1 f(\sqrt{a^2 + b^2 + c^2} w) (1 - w^2) dw,$$

其中 a, b, c 不全为零.

例4 求球面 $x^2 + y^2 + z^2 = R^2$ 被平面 $2x + 2y + z = R$ 所截两部分的面积.

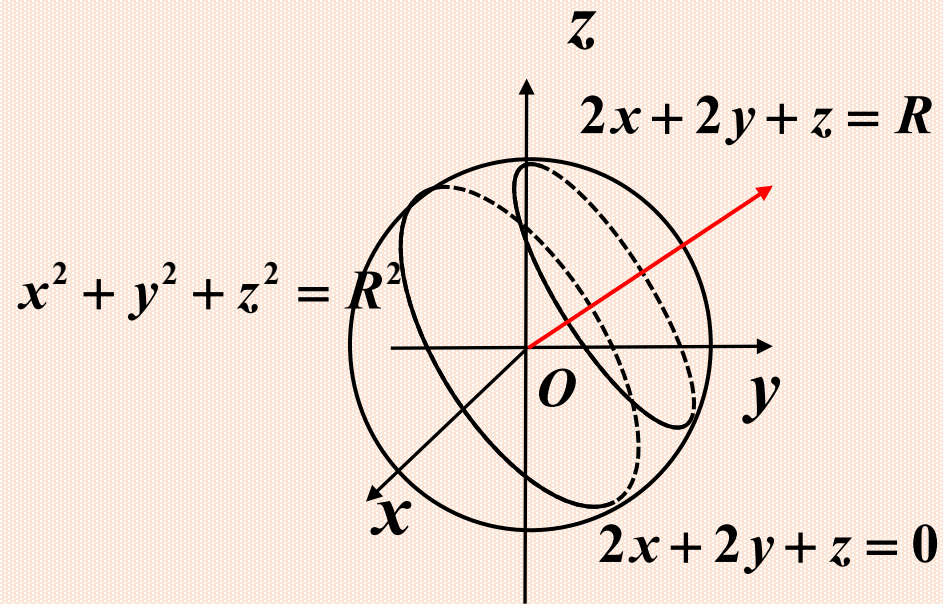
解 作旋转变换 $T : (x, y, z) \rightarrow (u, v, w)$, 使得在新坐标系 (u, v, w) 下

$2x + 2y + z = 0$ 对应 $w = 0$. 则 T 形如

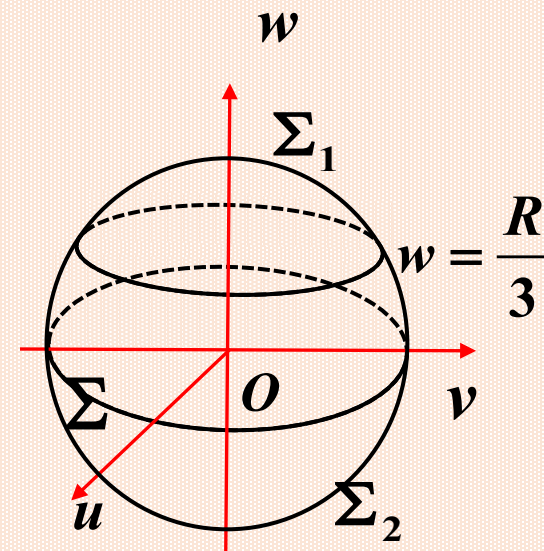
$$u = a_1 x + b_1 y + c_1 z,$$

$$v = a_2 x + b_2 y + c_2 z,$$

$$w = \frac{2x + 2y + z}{\sqrt{2^2 + 2^2 + 1^2}} = \frac{2x + 2y + z}{3}.$$



易知, 球面 $x^2 + y^2 + z^2 = R^2$ 在 T 的作用下旋转为 $\Sigma: u^2 + v^2 + w^2 = R^2$, 平面 $2x + 2y + z = R$ 在 T 的作用下旋转为 $\Pi: w = \frac{R}{3}$,



问题转化为求 Σ 被 Π 所截两部分 Σ_1, Σ_2 的面积, 其中,

$$\Sigma_1: w = \sqrt{R^2 - u^2 - v^2}, \quad (u, v) \in D = \left\{ (u, v) \mid u^2 + v^2 \leq \frac{8}{9} R^2 \right\}, \quad \Sigma_2 = \Sigma \setminus \Sigma_1,$$

Σ_1 的面积

$$S_1 = \iint_D \sqrt{1 + \left(\frac{\partial w}{\partial u} \right)^2 + \left(\frac{\partial w}{\partial v} \right)^2} du dv = \int_0^{2\pi} d\theta \int_0^{\frac{2\sqrt{2}}{3}R} \frac{R}{\sqrt{R^2 - r^2}} \cdot r dr = \frac{4\pi R^2}{3},$$

于是 Σ_2 的面积为
$$S_2 = 4\pi R^2 - \frac{4}{3}\pi R^2 = \frac{8}{3}\pi R^2.$$

例5 设 $f(u)$ 为连续函数, a, b, c 不全为 0, 证明

$$\iint_{x^2+y^2+z^2=1} f(ax+by+cz) dS = 2\pi \int_{-1}^1 f(\sqrt{a^2+b^2+c^2} u) du.$$

证明 作旋转变换, 使得

$$u = \frac{ax+by+cz}{\sqrt{a^2+b^2+c^2}}, v = v(x, y, z), w = w(x, y, z)$$

则 T 把球面 $x^2+y^2+z^2=1$ 映为球面 $u^2+v^2+w^2=1$,

记 $\lambda = \sqrt{a^2+b^2+c^2}$, 则有,

$$\begin{aligned}
\iint_{x^2+y^2+z^2=1} f(ax+by+cz) dS &= \iint_{u^2+v^2+w^2=1} f(\lambda u) dS \\
&= \iint_{\substack{\Sigma_1: u^2+v^2+w^2=1 \\ w \geq 0}} + \iint_{\substack{\Sigma_2: u^2+v^2+w^2=1 \\ w \leq 0}} f(\lambda u) dS \\
&= 2 \iint_{\Sigma_1} f(\lambda u) dS = 2 \int_{-1}^1 du \int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} \frac{f(\lambda u)}{\sqrt{1-u^2-v^2}} dv \\
&= 2 \int_{-1}^1 f(\lambda u) du \int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} \frac{1}{\sqrt{1-u^2}} \frac{dv}{\sqrt{1-\left(\frac{v}{\sqrt{1-u^2}}\right)^2}} \\
&= 2 \int_{-1}^1 f(\lambda u) \arcsin \frac{v}{\sqrt{1-u^2}} \bigg|_{v=-\sqrt{1-u^2}}^{v=\sqrt{1-u^2}} du = 2\pi \int_{-1}^1 f(\sqrt{a^2+b^2+c^2} u) du.
\end{aligned}$$

例6 设 $f(x, y, z) = \begin{cases} 1 - x^2 - y^2 - z^2 & \text{当 } x^2 + y^2 + z^2 \leq 1 \\ 0 & \text{当 } x^2 + y^2 + z^2 > 1 \end{cases}$

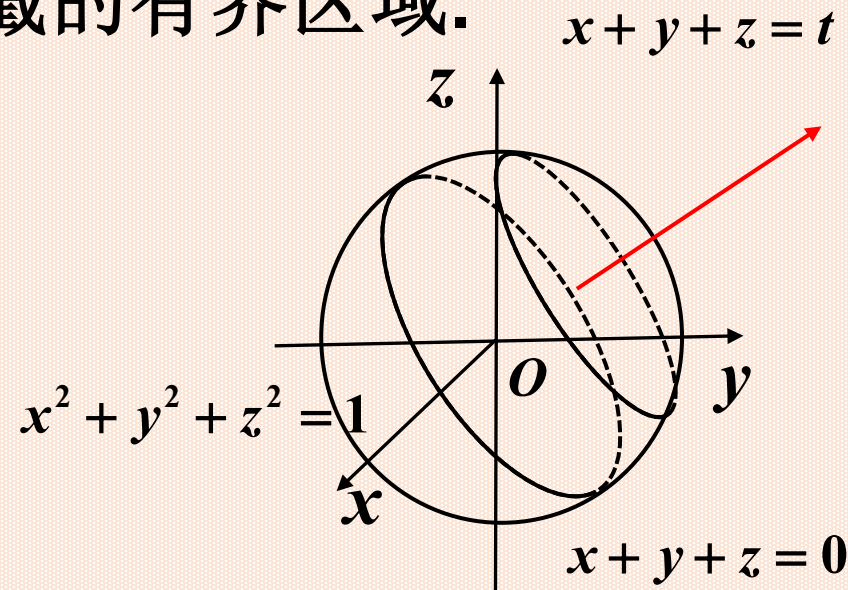
计算 $F(t) = \iint_{x+y+z=t} f(x, y, z) dS.$

解 $F(t) = \iint_{\Sigma} f(x, y, z) dS,$

其中 Σ 为 $x + y + z = t$ 被 $x^2 + y^2 + z^2 = 1$ 所截的有界区域.

作旋转变换,

$$w = \frac{x + y + z}{\sqrt{3}}, u = u(x, y, z), v = v(x, y, z)$$



则 T 把球面 $x^2 + y^2 + z^2 = 1$ 映为球面 $u^2 + v^2 + w^2 = 1$,

把平面 $x + y + z = t$ 映为平面 $w = \frac{t}{\sqrt{3}}$,

(1) 当 $t \notin [-\sqrt{3}, \sqrt{3}]$ 时, $\Sigma = \emptyset$, 即两曲面没有截出图形,
所以此时 $F(t) = 0$.

(2) 当 $t \in [-\sqrt{3}, \sqrt{3}]$ 时, $\Sigma: w = \frac{t}{\sqrt{3}}, (u, v) \in D_t$

其中, $D = \{(u, v) \mid u^2 + v^2 \leq 1 - \frac{t^2}{3}\}$

所以,

$$\begin{aligned} F(t) &= \iint_{D_t} \left(1 - \frac{t^2}{3} - u^2 - v^2\right) du dv = \int_0^{2\pi} d\theta \int_0^{\sqrt{1 - \frac{t^2}{3}}} \left(1 - \frac{t^2}{3} - r^2\right) r dr \\ &= \frac{\pi}{2} \left(1 - \frac{t^2}{3}\right)^2. \end{aligned}$$

